PHYS 365 — Final Exam Fall 2023 — Worked Solutions

30 questions. Every step uses the formula sheet printed at the back of the F2023 final. Anything that is *not* on that sheet is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key. Note: this sheet uses $\Delta x\,\Delta p_x \geq \dfrac{h}{2\pi}$.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	D	11	B	16	A	21	C	26	B
2	D	7	C	12	C	17	C	22	A	27	D
3	A	8	B	13	D	18	C	23	C	28	B
4	D	9	C	14	B	19	D	24	C	29	C
5	D	10	D	15	A	20	A	25	A	30	C

REMEMBER — not on the formula sheet

- SHM: $v = -\omega x_m \sin(\omega t + \phi)$ and $a = -\omega^2 x$. The slope of the $v(t)$ graph is a (Q1, Q2).
- Adding two waves with different amplitudes: $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\phi}$ (Q5).
- Transmitted wave on string 2 has speed v_2, so $k_2 = \dfrac{\omega}{v_2}$ (Q6).
- Line source: the sound spreads over a cylinder, $A = 2\pi r L$ (Q7).
- A pipe closed at **both** ends uses the same formula as one open at both ends: $f = nf_1$, $n = 1, 2, 3, \ldots$ (Q9).
- Small angles: $y_m = \dfrac{m\lambda D}{d}$ (bright, double slit) and $y_m = \dfrac{m\lambda D}{a}$ (dark, single slit) (Q10, Q15).
- Reflection from a moving car off a wall: do the Doppler shift twice (car to wall, then wall to car) (Q11).
- Springs in series: $k_{\text{eff}} = \dfrac{k_1 k_2}{k_1 + k_2}$, which is $\dfrac{k}{2}$ for two equal springs (Q18).
- Photon energy $E = \dfrac{hc}{\lambda}$. Number of photons per second $= \dfrac{P}{E_{photon}}$, with $P = IA$ (Q19, Q20, Q21).
- Compton: the electron gains the energy the photon loses, $K_e = E - E'$ (Q20).
- Infinite well state n has $n + 1$ nodes, counting both walls (Q23).
- Normalization: $\displaystyle\int |\psi|^2 dx = 1$ (Q25).

Question 1 Reading SHM from a velocity graph

1. The velocity $v(t)$ of a particle undergoing SHM is graphed in Figure 1. At point A, the particle momentarily is:

   A) stationary
   B) headed toward $x = -x_m$
   C) headed toward $x = +x_m$
   D) headed toward $x = 0$

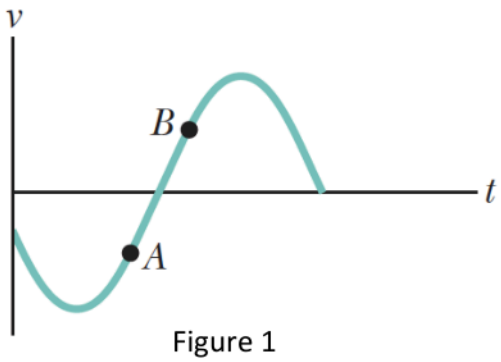


Figure 1

REMEMBER — not on the formula sheet

$a = \dfrac{dv}{dt}$ (the slope of the v graph), and $a = -\omega^2 x$.

Step 1. At point A, $v < 0$, so the particle is moving in the $-x$ direction.

Step 2. The slope at A is positive, so $a > 0$ and therefore $x < 0$. The particle is on the negative side, still moving toward $-x_m$.

Final answer: **(B) Headed toward $x = -x_m$** ✓ Matches answer key

Question 2 Phase constant from the velocity graph

2. What is the phase constant for the harmonic oscillator with the velocity function $v(t)$ given in Figure 2 if the position function $x(t)$ has the form $x = x_m \cos(\omega t + \phi)$? The vertical axis scale is set by v_s = 4.0 cm/s.

A) 4.07 rad
B) 3.25 rad
C) 0.93 rad
D) 5.36 rad

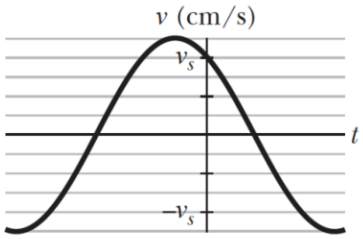


Figure 2

FROM THE FORMULA SHEET

$$x(t) = x_m \cos(\omega t + \Phi)$$

REMEMBER — not on the formula sheet

$$v = -\omega x_m \sin(\omega t + \phi) = -v_m \sin(\omega t + \phi), \quad a = -\omega^2 x_m \cos(\omega t + \phi)$$

Step 1. The gridlines are $1\ \frac{\text{cm}}{\text{s}}$ apart ($v_s = 4$ is 4 lines up). The peak is 5 lines up, so $v_m = 5\ \frac{\text{cm}}{\text{s}}$. At $t = 0$, $v = +4\ \frac{\text{cm}}{\text{s}}$ and the curve is going down.

Step 2. Solve for ϕ:

$$4 = -5 \sin\phi \ \Rightarrow\ \sin\phi = -0.8 \ \Rightarrow\ \phi = -0.927 \ \text{ or } \ \pi + 0.927 = 4.07 \text{ rad}$$

Step 3. v is decreasing, so $a(0) < 0$, which needs $\cos\phi > 0$. That selects $\phi = -0.927$ rad $= -0.927 + 2\pi = 5.36$ rad.

Final answer: **(D) 5.36 rad** ✓ **Matches answer key**

Question 3 Increasing the driving frequency

Questions 3, and 4 are related to the following situation:

You send a traveling wave along a particular string by oscillating one end.

3. If you increase the frequency of the oscillations, which statement is **true**?

A) The speed of the wave remains the same.
B) The speed of the wave increases.
C) The wavelength of the wave remains the same.
D) The wavelength of the wave increases.

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{\tau}{\mu}}, \quad v = f\lambda$$

Step 1. v depends only on τ and μ, so it stays the same. ($\lambda = \frac{v}{f}$ decreases.)

Final answer: **(A) Speed stays the same** ✓ **Matches answer key**

Question 4 Increasing the tension

4. If, you increase the tension in the string, which statement is **true**?

A) The speed of the wave decreases.
B) The speed of the wave remains the same.
C) The wavelength of the wave decreases.
D) The wavelength of the wave increases.

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{\tau}{\mu}}, \quad v = f\lambda$$

Step 1. A larger τ means a larger v. The frequency is fixed by the source, so $\lambda = \frac{v}{f}$ increases.

Final answer: **(D) Wavelength increases** ✓ Matches answer key

Question 5 Adding waves with a $\frac{\pi}{2}$ phase difference

5. Two sinusoidal waves of the same frequency travel in the same direction along a string. If y_{m1} = 3.0 cm, y_{m2} = 5.0 cm, ϕ_1 = 0, and $\phi_2 = \pi/2$ rad, what is the amplitude of the resultant wave?

A) 34 cm
B) 0
C) 15 cm
D) 5.8 cm

FROM THE FORMULA SHEET
$\tilde{A} = Ae^{i\theta}$

Step 1. The phasors are 3.0 cm along 0 and 5.0 cm along $\frac{\pi}{2}$, which are perpendicular:

$$A = \sqrt{3.0^2 + 5.0^2} = \sqrt{34} = 5.8 \text{ cm}$$

Final answer: **(D) 5.8 cm** ✓ Matches answer key

Question 6 Transmitted wave on a joined string

6. Two very long strings are joined at the position $x = 0$ and placed under a tension of 100 N. The string at $x < 0$ has mass density $\mu = 0.5$ kg/m while the string at $x > 0$ has mass density $\mu = 1$ kg/m. A wave of the form $y(x,t) = A\cos(kx - \omega t)$ is incident from $x < 0$. If the incident angular frequency ω is 1.0 rad/s and the incident amplitude A is 10 cm, then the displacement of the particle at $x = +1.0$ m is:

A) $y(t) = (8.3\text{ cm})\cos(\omega t - 0.63)$
B) $y(t) = (6.7\text{ cm})\cos(\omega t - 0.63)$
C) $y(t) = (6.7\text{ cm})\cos(\omega t - 0.10)$
D) $y(t) = (8.3\ cm)\cos(\omega t - 0.10)$

FROM THE FORMULA SHEET

$$Z = \sqrt{\mu\tau}, \quad T = \frac{2Z_1}{Z_1 + Z_2}, \quad v = \sqrt{\frac{\tau}{\mu}}, \quad v = \frac{\omega}{k}$$

Step 1. Impedances: $Z_1 = \sqrt{0.5(100)} = 7.07,\ Z_2 = \sqrt{1(100)} = 10$

$$T = \frac{2(7.07)}{7.07 + 10} = 0.828 \ \Rightarrow\ A_t = 0.828(10\text{ cm}) = 8.3\text{ cm}$$

Step 2. Speed and wavenumber on string 2:

$$v_2 = \sqrt{\frac{100}{1}} = 10\ \tfrac{\text{m}}{\text{s}}, \qquad k_2 = \frac{\omega}{v_2} = \frac{1.0}{10} = 0.10\ \tfrac{\text{rad}}{\text{m}}$$

Step 3. At $x = 1.0$ m: $\cos(0.10 - \omega t) = \cos(\omega t - 0.10)$ (cosine is even)

Final answer: **(D)** $(8.3\text{ cm})\cos(\omega t - 0.10)$ ✓ **Matches answer key**

Question 7 Intensity from a line source

7. In Figure 3, an electric spark jumps along a straight line of length L = 10 m, emitting a pulse of sound that travels radially outward from the spark. (The spark is said to be a line source of sound.) The power of this acoustic emission is P_s = 1.6 × 10^4 W. What is the intensity I of the sound when it reaches a distance r = 16 m from the spark?

A) 12 W/m^2
B) 10 W/m^2
C) 16 W/m^2
D) 21 W/m^2

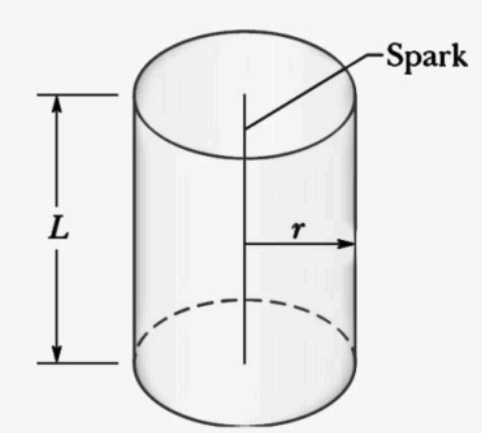


Figure 3

FROM THE FORMULA SHEET

$$I = \frac{P}{A}$$

REMEMBER — not on the formula sheet

The sound spreads out over the side of a cylinder: $A = 2\pi rL$.

$$I = \frac{P}{2\pi rL} = \frac{1.6 \times 10^4}{2\pi(16)(10)} = 15.9 \approx 16 \ \frac{\text{W}}{\text{m}^2}$$

Final answer: **(C) 16 W/m^2** ✓ Matches answer key

Question 8 Matching harmonics of two pipes

8. Pipe A, with length L, and pipe B, with length $2L$, both have two open ends. Which harmonic of pipe B has the same frequency as the first harmonic (fundamental frequency) of pipe A?

A) First
B) Second
C) Third
D) Fourth

FROM THE FORMULA SHEET
$f = \frac{nv}{2L}$ (open at both ends)

$$\frac{v}{2L} = \frac{nv}{2(2L)} \Rightarrow n = 2$$

Final answer: **(B) Second** ✓ Matches answer key

Question 9 Tube closed at both ends

9. For a particular tube with both ends closed, here are four of the six harmonic frequencies below 1000 Hz: 300 Hz, 600 Hz, 750 Hz, and 900 Hz. What two frequencies are missing from the list?

A) 100 Hz, and 200 Hz
B) 250 Hz, and 350 Hz
C) 150 Hz, and 450 Hz
D) 300 Hz, and 800 Hz

REMEMBER — not on the formula sheet
Closed at both ends works just like open at both ends: $f = nf_1, n = 1, 2, 3, \ldots$

Step 1. f_1 must divide every listed frequency: $\gcd(300, 600, 750, 900) = 150$ Hz.

Step 2. Harmonics below 1000 Hz: 150, 300, 450, 600, 750, 900, which is six of them. The missing two are 150 Hz and 450 Hz.

Final answer: **(C) 150 Hz and 450 Hz** ✓ Matches answer key

Question 10 Distance between $m = \pm 2$ fringes

10. Light from a HeNe laser (λ = 632.8 nm) is incident on 2 slits separated by 0.125 mm. Find the distance between the m = +2 and the m = −2 bright fringes on a screen 4.2 m away.

A) 75 mm
B) 65 mm
C) 43 mm
D) 85 mm

FROM THE FORMULA SHEET
$d \sin\theta = m\lambda$

REMEMBER — not on the formula sheet
Small angles: $y_m = \dfrac{m\lambda D}{d}$

$$y_{+2} - y_{-2} = \frac{4\lambda D}{d} = \frac{4(632.8 \times 10^{-9})(4.2)}{0.125 \times 10^{-3}} = 0.085 \text{ m} = 85 \text{ mm}$$

Final answer: **(D) 85 mm** ✓ Matches answer key

Question 11 Echo from a wall

11. A person on a railroad car blows a trumpet note at 440 Hz. The car is moving toward a wall at 20.0 m/s. What frequency does the trumpeter hear when the sound is reflected off the wall and back to the trumpeter?

A) 467 Hz
B) 494 Hz
C) 416 Hz
D) 440 Hz

FROM THE FORMULA SHEET

$$f' = f\,\frac{v \pm v_D}{v \pm v_S}, \qquad v_{sound} = 343\ \tfrac{\text{m}}{\text{s}}$$

REMEMBER — not on the formula sheet

Two steps: the wall hears a moving source, then the wall re-emits as a stationary source to a moving listener.

Step 1. Wall (stationary detector, source approaching): $f_w = f\,\dfrac{v}{v-u}$

Step 2. Trumpeter (detector approaching, stationary source): $f'' = f_w\,\dfrac{v+u}{v}$

$$f'' = f\,\frac{v+u}{v-u} = 440 \cdot \frac{343+20}{343-20} = 440 \cdot \frac{363}{323} = 494\ \text{Hz}$$

Final answer: **(B) 494 Hz** ✓ Matches answer key

Question 12 Guitar string tension

12. A guitar string has a mass per unit length of 0.5 g/m and a 65 cm length. What is the tension required to tune its fundamental frequency (first harmonic) to 330 Hz?

A) 74 N
B) 83 N
C) 92 N
D) 54 N

FROM THE FORMULA SHEET

$$f = \frac{nv}{2L'} \qquad v = \sqrt{\frac{\tau}{\mu}}$$

Step 1. Speed: $v = 2Lf = 2(0.65)(330) = 429\ \frac{\text{m}}{\text{s}}$

Step 2. Tension: $\tau = \mu v^2 = (0.5 \times 10^{-3})(429)^2 = 92$ N

Final answer: **(C) 92 N** ✓ Matches answer key

Question 13 Path difference from a step

13. Figure 4 shows two rays of light, of wavelength 600 nm, that reflect from glass surfaces separated by 150 nm. The rays are initially in phase. What is the path length difference of the rays?

A) 150 nm
B) 75 nm
C) 450 nm
D) 300 nm

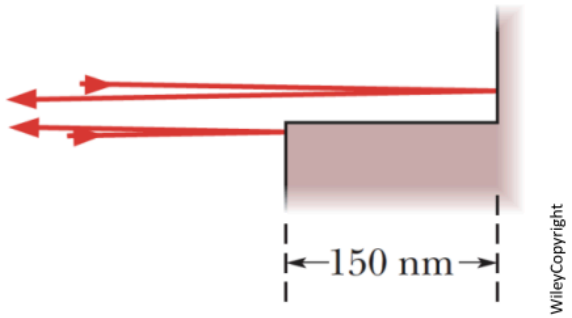


Figure 4

Step 1. The lower ray travels an extra 150 nm in and 150 nm back out:

$$\Delta L = 2(150) = 300 \text{ nm}$$

Final answer: **(D) 300 nm** ✓ Matches answer key

Question 14 Fraction of energy transmitted

14. In figure 5, a wave on a string is incident on a boundary between two strings. The wave travels from the left, on the string of linear mass density μ, to the right, onto the string of linear mass density $\alpha\mu$ (i.e., the linear mass density of the string on right is α times that of the string on the left). Which option gives the fraction of wave energy transmitted at this boundary?

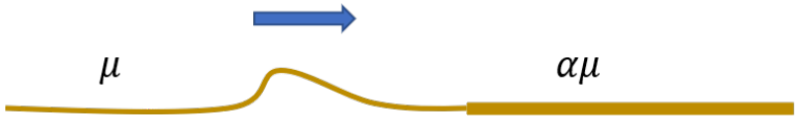


Figure 5

A) $\frac{2\sqrt{\alpha}}{1+\sqrt{\alpha}}$
B) $\frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$
C) $\frac{2}{1+\sqrt{\alpha}}$
D) $\frac{4\sqrt{\alpha}}{1+\sqrt{\alpha}}$

FROM THE FORMULA SHEET

$$Z = \sqrt{\mu\tau}, \quad T = \frac{2Z_1}{Z_1 + Z_2}, \quad T_e = \frac{Z_2}{Z_1}T^2$$

Step 1. Same tension: $\frac{Z_2}{Z_1} = \sqrt{\alpha}$, so $T = \frac{2}{1+\sqrt{\alpha}}$.

$$T_e = \sqrt{\alpha} \cdot \frac{4}{(1+\sqrt{\alpha})^2} = \frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$$

Final answer: **(B)** $\frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$ ✓ Matches answer key

Question 15 Spacing of single-slit minima

15. A slit 1.00 mm wide is illuminated by light of wavelength 589 nm. We see a diffraction pattern on a screen 3.00 m away. What is the distance between the first two diffraction minima on the same side of the central diffraction maximum?

A) 1.77 mm
B) 3.53 mm
C) 1.18 mm
D) 5.89 mm

FROM THE FORMULA SHEET
$a \sin\theta = m\lambda$ (minima)

REMEMBER — not on the formula sheet
Small angles: $y_m = \dfrac{m\lambda D}{a}$, so neighbouring minima are $\dfrac{\lambda D}{a}$ apart.

$$\Delta y = \frac{\lambda D}{a} = \frac{(589 \times 10^{-9})(3.00)}{1.00 \times 10^{-3}} = 1.77 \times 10^{-3}\ \text{m} = 1.77\ \text{mm}$$

Final answer: **(A) 1.77 mm** ✓ Matches answer key

Question 16 Path difference to a point on the screen

Questions 16 and 17 are related to the following situation:

In the Figure 6, two isotropic point sources S_1 and S_2 emit light in phase at wavelength λ and at the same amplitude. The sources are separated by distance $d = 10\,\lambda$ on an x axis. A viewing screen is at distance $D = 20\,\lambda$ from S_2 and parallel to the y axis. The figure shows two rays reaching point P on the screen, at height y_P.

16. If $y_p = d$, what is the path length difference between the two rays?

A) 9.3 λ
B) 5.8 λ
C) 7.5 λ
D) 6.4 λ

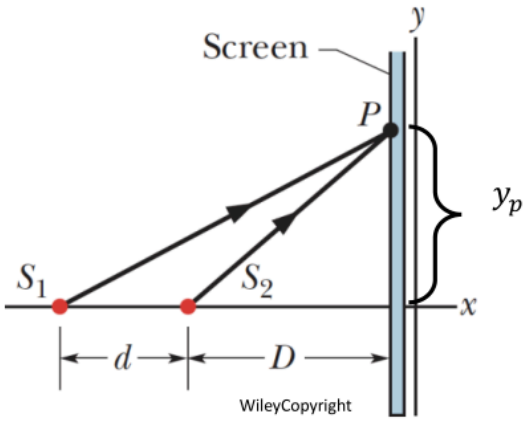


Figure 6

Step 1. Coordinates in units of λ: P is at height 10, S_2 is 20 back horizontally, and S_1 is 30 back.

$$r_1 = \sqrt{30^2 + 10^2} = 31.62\lambda, \qquad r_2 = \sqrt{20^2 + 10^2} = 22.36\lambda$$

$$\Delta L = 31.62 - 22.36 = 9.3\lambda$$

Final answer: **(A) 9.3 λ** ✓ Matches answer key

Question 17 Intensity for a 7.8λ difference

17. If the path length difference between two rays from S_1 and S_2 at P (above figure) equals to 7.8 λ, the resulting intensity at point P is:

A) Minimum
B) Maximum
C) Intermediate but closer to maximum
D) Path length difference 7.8 λ for the above situation is impossible.

FROM THE FORMULA SHEET

Constructive: $\frac{\Delta L}{\lambda} = 0, 1, 2, \ldots$ Destructive: $\frac{\Delta L}{\lambda} = 0.5, 1.5, \ldots$

Step 1. The largest possible ΔL is $d = 10\lambda$ (on the axis), so 7.8λ is possible.

Step 2. 7.8 is 0.2 away from 8 (a maximum) and 0.3 away from 7.5 (a minimum), so the intensity is intermediate but closer to a maximum.

Final answer: **(C) Intermediate, closer to maximum** ✓ Matches answer key

Question 18 Two springs in series

18. In Figure 7 , two springs are joined and connected to a block of mass 0.245 kg that is set oscillating over a frictionless floor. The springs each have spring constant k = 6430 N/m. What is the frequency of the oscillations?

A) 75 Hz
B) 54 Hz
C) 18 Hz
D) 38 Hz

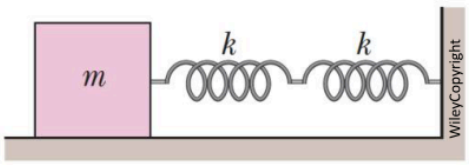


Figure 7

FROM THE FORMULA SHEET

$$\omega = \sqrt{\frac{k}{m}}, \quad \omega = 2\pi f$$

REMEMBER — not on the formula sheet

Springs in series: $k_{\text{eff}} = \dfrac{k}{2}$ for two identical springs.

$$k_{\text{eff}} = \frac{6430}{2} = 3215\ \tfrac{\text{N}}{\text{m}}, \qquad f = \frac{1}{2\pi}\sqrt{\frac{3215}{0.245}} = \frac{114.6}{2\pi} = 18\ \text{Hz}$$

Final answer: **(C) 18 Hz** ✓ Matches answer key

Question 19 Photons absorbed per second

19. A solar panel on a satellite having a surface area of 2.00 m^2 absorbs solar energy having an irradiance of 1360 W/m^2 and a wavelength of 500 nm. How many photons are absorbed each second, assuming 100% absorption?

A) 1.71×10^{21}
B) 3.42×10^{21}
C) 1.37×10^{22}
D) 6.84×10^{21}

FROM THE FORMULA SHEET

$$E = hf, \quad I = \frac{P}{A}$$

REMEMBER — not on the formula sheet

$E_{photon} = \dfrac{hc}{\lambda}$. Photons per second $= \dfrac{P}{E_{photon}}$.

Step 1. Power: $P = IA = 1360(2.00) = 2720$ W

Step 2. Energy per photon:

$$E = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{500 \times 10^{-9}} = 3.98 \times 10^{-19}\ \text{J}$$

Step 3. Photon rate: $N = \dfrac{2720}{3.98 \times 10^{-19}} = 6.84 \times 10^{21}$ per second

Final answer: **(D)** 6.84×10^{21} ✓ Matches answer key

Question 20 Compton angle from the electron's energy

20. In the bright and sunny Physics Lab #39458, photons are incident on a target containing stationary electrons. The incident radiation has a wavelength of 2.89×10^{-12} m, and the detector is placed at an angle of φ from the initial direction of the photons. An electron from which one of these photons was scattered receives an energy of 120.6 keV. What is the angle φ? (Answer in degrees, with $0 \leq \varphi \leq 180°$).

A) 57.6°
B) 62.4°
C) 122.4°
D) 117.6°

FROM THE FORMULA SHEET

$\Delta\lambda = \dfrac{h}{mc}(1 - \cos\phi), \quad E = hf$

REMEMBER — not on the formula sheet

$E = \dfrac{hc}{\lambda}$. Energy conservation: $E' = E - K_e$.

Step 1. Incident photon energy:

$$E = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{2.89 \times 10^{-12}} = 6.88 \times 10^{-14}\ \text{J} = 429.6\ \text{keV}$$

Step 2. Scattered photon: $E' = 429.6 - 120.6 = 309.0$ keV

$$\lambda' = \frac{hc}{E'} = 4.018 \times 10^{-12}\ \text{m}, \qquad \Delta\lambda = 4.018 - 2.89 = 1.128\ \text{pm}$$

Step 3. Angle, with $\dfrac{h}{m_e c} = 2.426$ pm:

$$1 - \cos\phi = \frac{1.128}{2.426} = 0.465 \;\Rightarrow\; \cos\phi = 0.535 \;\Rightarrow\; \phi = 57.6^\circ$$

Final answer: **(A) 57.6°** ✓ Matches answer key

Question 21 Comparing laser wavelengths

21. You perform a pair of photoelectric experiments:

- Laser 1 is incident on surface 1, with work function Φ_1 = 2 eV.
- Laser 2 is incident on surface 2, with work function Φ_2 = 4 eV .

It is found that, in both cases, electrons with a maximum kinetic energy of 2 eV are ejected. How do the wavelengths of the lasers compare?

A) $\lambda_2 = \frac{1}{2}\lambda_1$
B) $\lambda_2 = \frac{3}{2}\lambda_1$
C) $\lambda_2 = \frac{2}{3}\lambda_1$
D) $\lambda_2 = 2\lambda_1$

FROM THE FORMULA SHEET
$hf = K_{max} + \Phi$

REMEMBER — not on the formula sheet
$hf = \frac{hc}{\lambda}$, so $\lambda \propto \frac{1}{E}$.

Step 1. Photon energies: $E_1 = 2 + 2 = 4$ eV and $E_2 = 2 + 4 = 6$ eV.

$$\frac{\lambda_2}{\lambda_1} = \frac{E_1}{E_2} = \frac{4}{6} = \frac{2}{3}$$

Final answer: **(C)** $\lambda_2 = \frac{2}{3}\lambda_1$ ✓ Matches answer key

Question 22 Ranking tunneling probabilities

22. In an experimental set-up, three quantum tunneling conditions are tested. Electrons with energy E approach the following potential energy barriers. A detector placed on the other side measures if the electrons are transmitted.

- Barrier (1) has height of $10E$ and thickness of L.
- Barrier (2) has height of $17E$ and thickness of $L/2$.
- Barrier (3) has height of $2E$ and thickness of $2L$.

Rank the barriers according to the probability of the electron tunneling through the barrier, from the largest probability to smallest.

A) (2) = (3) > (1)
B) (1) > (2) = (3)
C) (1) > (3) > (2)
D) (2) > (3) > (1)

FROM THE FORMULA SHEET

$$T \approx e^{-2bL}, \quad b = \sqrt{\frac{8\pi^2 m(U_b - E)}{h^2}}$$

Step 1. $b \propto \sqrt{U_b - E}$, so compare $bL \propto \sqrt{U_b - E} \times \text{thickness}$:

$$(1) : \sqrt{9E}\, L = 3L\sqrt{E}, \qquad (2) : \sqrt{16E}\, \frac{L}{2} = 2L\sqrt{E}, \qquad (3) : \sqrt{E}\,(2L) = 2L\sqrt{E}$$

Step 2. A smaller bL means a larger T: (2) = (3) > (1).

Final answer: **(A) (2) = (3) > (1)** ✓ **Matches answer key**

Question 23 Counting nodes

23. An infinite, one-dimensional potential well of width L contains a single electron of energy E. In repeated experiments it is found that there are four places within the well (including the two endpoints) where the electron is never detected. What is the electron energy E in terms of the ground state energy E_1?

A) E_1
B) 4 E_1
C) 9 E_1
D) 16 E_1

FROM THE FORMULA SHEET

$$\psi_n = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{h^2}{8mL^2}n^2$$

REMEMBER — not on the formula sheet

ψ_n has $n + 1$ zeros, counting both walls.

Step 1. Four zeros means $n = 3$, so $E = 3^2 E_1 = 9E_1$.

Final answer: **(C)** $9E_1$ ✓ Matches answer key

Question 24 Doubling the well width

24. A one-dimensional infinite potential well in the region $0 \leq x \leq L$ has allowed electron energy levels of 1, 4, 9, 16, ... eV. If the width of the well is slowly doubled to $2L$, what is the new energy of the $n = 4$ level?

A) 1 eV
B) 2 eV
C) 4 eV
D) 8 eV

FROM THE FORMULA SHEET

$$E_n = \frac{h^2}{8mL^2}n^2$$

Step 1. Original $n = 4$ level: $E_4 = 16$ eV. Since $E \propto \frac{1}{L^2}$, doubling L divides it by 4:

$$E_4' = \frac{16}{4} = 4 \text{ eV}$$

Final answer: **(C) 4 eV** ✓ **Matches answer key**

Question 25 Normalizing $A\sqrt{x}$

25. A wavefunction describing a quantum particle in one dimension is given by

$$\psi(x) = \begin{cases} A\sqrt{x}, & \text{for } 0 \le x \le L \\ 0, & \text{for all other } x \end{cases}$$

If $L = 0.78$ mm, what value of A is needed to normalize this wavefunction?

A) $1.81\ \text{mm}^{-1}$

B) $2.56\ \text{mm}^{-1}$

C) $2.18\ \text{mm}^{-1}$

D) $0.30\ \text{mm}^{-1}$

REMEMBER — not on the formula sheet

Normalization: $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$

$$\int_0^L A^2 x\, dx = A^2 \frac{L^2}{2} = 1 \;\Rightarrow\; A = \frac{\sqrt{2}}{L} = \frac{1.414}{0.78\ \text{mm}} = 1.81\ \text{mm}^{-1}$$

Final answer: **(A) 1.81 mm^{-1}** ✓ Matches answer key

Question 26 Longest emitted wavelength from $n = 3$

26. An electron is in a one-dimensional infinite potential well of length L = 104 pm. The electron has ground state energy E_1. What is the longest wavelength photon that could be emitted by the system in a transition from the energy level $n = 3$?

A) 5.10 nm
B) 7.13 nm
C) 11.9 nm
D) 4.45 nm

FROM THE FORMULA SHEET

$$E_n = \frac{h^2}{8mL^2}n^2, \quad E = hf$$

Step 1. The smallest drop is $3 \to 2$: $\Delta E = (9-4)E_1 = 5E_1$.

$$E_1 = \frac{(6.63\times10^{-34})^2}{8(9.109\times10^{-31})(104\times10^{-12})^2} = 5.58\times10^{-18}\ \text{J}$$

$$\lambda = \frac{hc}{5E_1} = \frac{1.989\times10^{-25}}{2.79\times10^{-17}} = 7.13\times10^{-9}\ \text{m}$$

Final answer: **(B) 7.13 nm** ✓ Matches answer key

Question 27 Finding n_{low}

27. The energy levels of a hydrogen-like quantum system (with one electron bound to a potential well) are found to be

$$E_n = -2.6\text{eV}/n^2.$$

As the electron makes a transition from a bound state with quantum number n_{high} to a bound state with quantum number n_{low}, you observe a photon of wavelength 637 nm emitted from this system. What is n_{low}?

A) 2
B) 3
C) 4
D) 1

FROM THE FORMULA SHEET

$E = hf$

REMEMBER — not on the formula sheet

$E_{photon} = \dfrac{hc}{\lambda}$, with $hc = 1240$ eV·nm

Step 1. Photon energy: $E = \dfrac{1240}{637} = 1.947$ eV

$$2.6\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right) = 1.947 \;\Rightarrow\; \frac{1}{n_{low}^2} - \frac{1}{n_{high}^2} = 0.749$$

Step 2. Only $n_{low} = 1, n_{high} = 2$ works: $1 - \frac{1}{4} = 0.75$ ✓

Final answer: **(D)** $n_{low} = 1$ ✓ **Matches answer key**

Question 28 Uncertainty with a huge Planck constant

28. Imagine playing baseball in a universe (not ours!) where the Planck constant is 0.40 J·s. What would be the uncertainty in the position of a 0.40 kg baseball that is moving at 25 m/s along an axis if the uncertainty in the speed is 1.5 m/s?

A) 0.60 m
B) 0.11 m
C) 0.15 m
D) 0.30 m

FROM THE FORMULA SHEET

$$\Delta x\,\Delta p_x \geq \frac{h}{2\pi}$$

REMEMBER — not on the formula sheet

$$\Delta p = m\Delta v$$

$$\Delta x = \frac{h}{2\pi m \Delta v} = \frac{0.40}{2\pi(0.40)(1.5)} = 0.11\ \text{m}$$

Final answer: **(B) 0.11 m** ✓ Matches answer key

Question 29 Compton-scattered wavelength

29. Light of wavelength 2.40 pm is directed onto a target containing free electrons. Find the wavelength of light scattered at 30° from the incident direction. The electron Compton wavelength is 2.43×10^{-12} m.

A) 3.65 pm
B) 6.04 pm
C) 2.73 pm
D) 4.32 pm

FROM THE FORMULA SHEET

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$$

$$\lambda' = 2.40 + 2.43(1 - \cos 30^\circ) = 2.40 + 0.33 = 2.73 \text{ pm}$$

Final answer: **(C) 2.73 pm** ✓ Matches answer key

Question 30 Ranking energy differences

30. Rank the following pairs of quantum states for an electron confined to an infinite well according to the energy differences between the states, greatest first: (a) n = 3 and n = 1, (b) n = 5 and n = 4, (c) n = 4 and n = 3.

A) (a) > (b) > (c)
B) (b) > (c) > (a)
C) (b) > (a) > (c)
D) (a) > (c) > (b)

FROM THE FORMULA SHEET

$$E_n \propto n^2$$

$$(a) : 9 - 1 = 8, \qquad (b) : 25 - 16 = 9, \qquad (c) : 16 - 9 = 7$$

Final answer: **(C) (b) > (a) > (c)** ✓ Matches answer key